

$$I = \{x | (x \in A \wedge (x \notin C \wedge x \in C)) \wedge (x \notin B)\} \quad (\text{By 2nd Associative Law of Propositions})$$

$$I = \{x | (x \in A \wedge (x \in C \wedge x \notin C)) \wedge (x \notin B)\} \quad (\text{2nd Commutative Law of Propositions})$$

$$= \{x | (x \in A \wedge (x \in C \wedge x \in \bar{C})) \wedge (x \notin B)\} \quad (\text{Defn. 5})$$

$$= \{x | (x \in A \wedge (x \in C \wedge \bar{C})) \wedge (x \notin B)\} \quad (\text{Defn. 2})$$

$$= \{x | (x \in A \wedge x \in \emptyset) \wedge (x \notin B)\} \quad (\text{Prob 9-b})$$

$$(x \notin B) = \{x | x \notin A \wedge \phi \wedge x \notin B\} \quad (\text{Defn. 2})$$

$$= \{x | x \notin \phi \wedge x \notin B\} \quad (\text{Prob 7-b})$$

$$= \{x | x \notin B \wedge x \notin \phi\} \quad (\text{2nd Commutative Law of Propositions})$$

$$= \{x | x \notin B \wedge x \in \emptyset\} \quad (\text{Defn. 5})$$

$$= \{x | x \notin B \wedge \phi\} \quad (\text{Defn. 2}) = \{x | x \in \emptyset\} \quad (\text{Prob 7-b}) = \emptyset$$

$$e) (B-A) \cup (C-A) = (B \cup C) - A$$

$$(B-A) \cup (C-A) = \{x | x \in (B-A) \vee x \in (C-A)\} \quad (\text{Defn. 1})$$

$$= \{x | (x \in B \wedge x \notin A) \vee (x \in C \wedge x \notin A)\} \quad (\text{Defn. 4})$$

$$= \{x | (x \notin A \wedge x \in B) \vee (x \notin A \wedge x \in C)\} \quad (\text{2nd Commutative Law of Propositions})$$

$$= \{x | (x \notin A \wedge (x \in B \vee x \in C))\} \quad (\text{2nd Distributive Law of Propositions})$$

$$= \{x | (x \notin A \wedge x \in B \cup C)\} \quad (\text{Defn. 1})$$

$$= \{x | x \in B \cup C \wedge x \notin A\} \quad (\text{2nd Commutative Law of Propositions})$$

$$= (B \cup C) - A \quad (\text{Defn. 4}) \quad \square$$

Let 2) Show that if A and B are sets, then

$$a) A - B = A \cap \bar{B} \quad (\text{By Defn. 4})$$

$$\text{Solution: } A - B = \{x | x \in A \wedge x \notin B\} \quad (\text{Defn. 5})$$

$$= A \cap \bar{B} \quad (\text{Defn. 2}) \quad \square$$

$$b) (A \cap B) \cup (A \cap \bar{B}) = A \quad (\text{Example 12})$$

$$\text{Soln: } (A \cap B) \cup (A \cap \bar{B}) = A \cap (B \cup \bar{B}) \quad (\text{Prob 9(a)})$$

$$= A \cap U \quad (\text{Prob 6(b)}) \quad \square$$